

# CHARAN SAI GURRALA

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## PROFESSIONAL SUMMARY

- 4 years as a Data Engineer, specializing in designing, developing, and optimizing data pipelines across Azure, AWS, and Snowflake.
- Expertise in data warehousing, data modeling, and ETL processes using tools like Azure Data Factory, Databricks, AWS Glue, and Redshift.
- Developed and optimized large-scale data processing pipelines using **Scala** and big data frameworks like **Apache Spark** and **Hadoop**.
- Leveraged **Scala's functional programming features** to implement efficient, high-performance data transformations and real-time analytics for distributed systems.
- Implemented high-throughput, low-latency NoSQL solutions using **Scylla**, optimizing database performance for distributed systems. Built unified data processing pipelines with **Apache Beam**, ensuring compatibility with multiple execution engines like Dataflow and Flink.
- Proficient in data manipulation, programming, and integrating new software systems to drive business value.
- Provisioned and managed GCP resources using Terraform to enable Infrastructure as Code (IaC) practices. Deployed and managed Apache Spark and Hadoop clusters using Google Cloud Dataproc for big data processing.
- Developed distributed data processing workflows using MapReduce to handle large-scale datasets efficiently.
- Strong knowledge of Kubernetes and Docker, enabling containerized deployment and orchestration of applications for scalable data engineering workflows. Expertise in designing and optimizing Teradata database schemas for large-scale data warehousing and analytics solutions.
- Automated repetitive workflows using Python and shell scripting, accelerating development sprints and improving delivery timelines.
- Managed release processes, coordinating with cross-functional teams to ensure timely deployment of updates while maintaining high-quality standards. Utilized strong communication skills to collaborate with stakeholders and streamline business processes, aligning data insights with operational improvements.

## TECHNICAL SKILLS

- **Cloud Technologies:** Snowflake, Azure(Blob Storage, Data Factory, Azure Synapse Analytics), AWS(EC2, S3, ECR, ECS, IAM, EMR), DBT, Databricks.
- **Languages:** Java, Python, HTML5, CSS, JavaScript, PHP, XML, Shell Scripting, C++.
- **Data Warehousing/ETL:** Informatica, Attunity, Fivetran.
- **Databases:** Oracle, IBM DB2, Sybase, SQL Server, MySQL & R.
- **Reporting Tools:** Power BI, Tableau & Looker.
- **Big Data Technologies:** HDFS, Map Reduce, Spark, Hive, Sqoop, Flume, Pyspark.
- **Data Modeling:** Star Schema Modeling, Snowflake Modeling, Data Modeler.
- **CI/CD Tools:** Confluence, Bit Bucket, Jenkins, Kubernetes, Kafka.
- **Other tools & Utilities:** SQL\*Plus, NoSQL, MongoDB, PostgreSQL, Jira, Putty, LINUX, Airflow, Snowpark, SSIS, SSAS, and GIT.

## PROFESSIONAL EXPERIENCE

Cardinal Health | Chicago, IL

June 2023 - Present

### Data Engineer

- Implemented data warehouse solutions and scalable **ETL pipelines** using **Azure Data Factory** and **Databricks**.
- Designed and implemented scalable data solutions using **AWS services**, including AWS Redshift, AWS Glue, and AWS Lambda, to support complex data processing and analytics needs.
- Troubleshoot and resolve **data integration** issues, providing operational support and adhering to SLAs and KPIs.
- Implemented predictive maintenance algorithms using Python and SageMaker, resulting in a 20% reduction in equipment downtime and maintenance costs.
- Implemented and optimized **healthcare** data integration solutions using InterSystems, enabling seamless data exchange between diverse healthcare systems, which enhanced patient data interoperability and reduced data transfer errors by 25%.
- Led the development of data pipelines and key tables for analytics within the data warehouse, leveraging DBT, while collaborating closely with the wider team.
- Applied **Agile, Scrum, and Kanban methodologies**, streamlining software development lifecycles and improving project execution.
- Proficient in integrating and managing data workflows with SAP, optimizing ERP systems for reporting and analytics. Experienced in designing and querying in-memory databases with SAP HANA for real-time data processing and analytics.
- Extracted, transformed, and loaded data into **CSV files** using **Python** and **SQL**, automating data processing and reducing manual efforts.
- Configured and managed **enterprise-level Snowflake environments**, ensuring optimal performance for data storage and processing workloads.
- Generated **ad-hoc Tableau reports** and visualizations from multiple data sources (Excel, CSV) to provide actionable insights to stakeholders.
- Skilled in containerizing applications using **Docker** and orchestrating them with **Kubernetes** for scalable deployments.
- Leveraged Unix utilities for file manipulation, process management, and system optimization in Linux-based environments.

**Tech Stack:** AWS (EC2, Lambda, Redshift, Glue, S3), Snowflake, SQL, Python, SSIS, T-SQL, Power BI, Erwin Data Modeler, MDM (Match/Merge), Data Cataloging (AWS Glue), Agile, Scrum, Logical and Physical Data Models, Metadata Design, Topic Modeling, Cosine Similarity, Text Preprocessing (Stopwords, Lemmatization)

## JP Morgan Chase

June 2021 - July 2022

### Data Engineer | New York, US

- Designed and maintained scalable ETL pipelines using **Apache Spark**, Python, Elasticsearch and SQL. Conducted performance tuning for SQL, NoSQL databases, and **Google BigQuery**, cutting query response times by 30%.
- Implemented workflows with AWS Glue, Amazon Redshift, and Azure Synapse, optimizing data transformation and reducing processing time by 40%.
- Built large-scale applications and data structures for data ingestion from Neo4j, MongoDB (Graph, NoSQL OLTP) databases in Airflow.
- Enhanced data analytics capabilities by implementing Databricks on AWS, contributing to increased accuracy and speed in data analysis, resulting in a 20% increase in data analysis speed and accuracy.
- Deployed machine learning models using **AWS SageMaker** and **Apache Airflow**, boosting predictive analytics accuracy by 25%.
- Led legacy system migration to cloud platforms with AWS DMS, Alteryx, Azure Data Factory, and Google BigQuery, ensuring data integrity.
- Experienced in managing dependencies and packages using **Composer**, ensuring efficient software builds.
- Implemented and managed data analytics workflows using **Microsoft Fabric** for unified data integration and insights. Designed and deployed CI/CD pipelines using Jenkins to automate build, test, and deployment processes.
- Built and managed scalable machine learning workflows with Kube Flow for streamlined MLOps.
- Deployed high-performance AI models using NVIDIA Triton Inference Server to enable efficient inference at scale. Monitored and managed machine learning lifecycle, including experimentation, reproducibility, and deployment, using MLflow.
- Automated data integration workflows using **ETL tools** such as **AWS Glue** and **Informatica**, enhancing data pipeline efficiency and reliability.
- Incorporated **REST APIs** to integrate external data sources into reporting platforms, expanding data availability and enriching analytics capabilities.

**Tech Stack:** Power BI Desktop, Power Query, DAX, KPI dashboards, Tableau, AWS Glue, Informatica, Google Cloud Platform (GCP), AWS, REST APIs, GCP Big Query, Snowflake, KPI dashboards, scatter plots, Data Blending, Cross-Database Joins, Indexing.

## Walmart

Jan 2020 - May 2021

### Data Engineer | New York, US

- Designed and implemented scalable data warehouses using Azure Synapse Analytics and Azure SQL Data Warehouse for enhanced inventory management and sales performance tracking.
- Monitored and managed application and infrastructure performance with Google Cloud Stackdriver for logging and observability. Designed and implemented low-latency NoSQL solutions using Google Cloud Bigtable for high-throughput applications.
- Played a key role in the **full software development lifecycle (SDLC)**, from **data analysis** through **implementation and support**.
- Configured **Row-level security** and **user filters** in Tableau to control **user access**, ensuring secure and tailored data views for different stakeholders.
- Skilled in leveraging **business analytics** to evaluate data-driven insights, improving decision-making processes and operational efficiency.
- Proficient in using **version control systems** like Git, TFS, and SVN to manage codebases, facilitate collaboration, and ensure code integrity.
- Experienced in using **data-related tools and technologies** like Power BI, SQL Server, and ETL pipelines to manage and analyze large datasets.
- Developed complex reports and dashboards in **MicroStrategy**, **PDF**, **Active HTML**, and **Excel**, delivering insights in various formats to meet business needs.
- Worked with **Power BI Paginated Reports** using **Report Builder** to deliver formatted, printable reports for stakeholders.
- Utilized **Teradata**, **Oracle**, **SQL**, **SSIS**, **SSRS**, **DOMO**, and **Snowflake** for efficient data processing, integration, and reporting.
- Leveraged **MS Excel** for data manipulation and analysis, supporting ad-hoc reporting and decision-making processes.

**Tech Stack:** Azure Synapse, Azure SQL Data Warehouse, GCP, Tableau, Power BI, MicroStrategy, SSRS, Teradata, Oracle, Snowflake, SQL, SSIS, DOMO, SharePoint, MS Excel.

## EDUCATION DETAILS

- **Western Illinois University** | Illinois, United States  
- **Masters** in Computer Science

August 2022 - May 2024

**Coursework:** Artificial Intelligence Methods, Data Structures & Algorithms, Core Java, Object Oriented Programming, Database Design and Administration, Information Technology, Ruby, Information Security, Advanced Artificial Intelligence (GEN AI), Information Science.

## CERTIFICATIONS

- AWS Solution Architect - Associate